University of Ottawa

Department of Mathematics and Statistics

MAT 1302C: Mathematical Methods II

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Final Exam – Solutions

December 2018

Surname _____ First Name _____

Student # _____ Seat # _____

Instructions:

(a) You have 3 hours to complete this exam.

(b) This exam consists of two parts. Questions 1 through 14 are *answer only*. For these questions, only your final answer will be considered for marks. Questions 15 through 21 are *long answer*. For these questions, you must show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.

(c) The number of points available for each question is indicated in square brackets.

(d) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this *clearly*. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(e) Write your student number at the top of each page in the space provided.

(f) You may use the second-to-last page of the exam as extra space for solutions. If you do so, indicate clearly on the page of the relevant question that you have continued your solution on the extra page.

(g) You may use the last page of the exam as scrap paper.

(h) Cellular phones, unauthorized electronic devices or course notes are not allowed during this exam. Phones and devices must be turned off and put away in your bag. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

By signing below, you acknowledge that you are complying with statement (h) above:

Signature: _____

Good luck!

Please do not write in the table below.

Question	1–14	15	16	17	18	19	20	21	Total
Maximum	24	4	4	4	4	6	4	5	55
Grade									

Part A: Answer Only Questions

For Questions 1–14, only your final answer will be considered for marks. Write your final answers in the spaces provided.

1. **[2 points]** Consider the matrices

$$A = \begin{bmatrix} -2 & 1 \\ -3 & 2 \\ 2 & -\frac{1}{2} \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -3 \\ 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 3 & 5 \end{bmatrix}$$

Compute $2A^T - B^TC$.

Answer: $2A^T - B^TC = \begin{bmatrix} -6 & -4 & 0 \\ 7 & -2 & 0 \end{bmatrix}$

2. **[2 points]** Suppose that A is an $m \times n$ matrix such that $n > m$ and $\text{rank}(A) < m$. For each statement below, write '**T**' if the statement is true, and write '**F**' if the statement is false. You will receive 0.5 points for each correct answer, lose 0.25 points for each incorrect answer, and receive zero points for an answer left blank. You cannot receive a negative score on this question.

F For every vector $\mathbf{b} \in \mathbb{R}^m$, the linear system $A\mathbf{x} = \mathbf{b}$ is consistent.

F The span of the columns of A is equal to $\mathbb{R}^m$.

T The reduced echelon form of A has at least one row whose entries are all zeros.

F $\text{Nul}(A) = \{\mathbf{0}\}$.

3. **[1 point]** Suppose that A, B, and C are 3×3 matrices such that $\det(A) = 2$, $\det(B) = 3$, and $\det(-2B^{-1}C^TA^2) = 16$. Calculate $\det(C)$.

Answer: $\det(C) = -\frac{3}{2}$.

4. **[1 point]** Determine all values of $t \in \mathbb{R}$ such that the linear system

$$\begin{cases} x_1 - 3x_2 + x_3 + 5x_4 &= 7 \\ 3x_2 + tx_3 - x_4 &= 1 \\ -6x_2 + 4x_3 + 2x_4 &= 5 \end{cases}$$

does not have any solutions.

Answer: $t = -2$

5. **[2 points]** Let $z = -1 - 2i$ and $w = 2 - 4i$. Write the complex number $\dfrac{i|w|^2}{w + \overline{z}}$ in the form $a + bi$ where a and b are real numbers.

Answer: $-8 + 4i$

6. **[2 points]** Let $A = \begin{bmatrix} 5 & -2 & -\frac{1}{3} & 8 \\ 0 & -5 & 3 & -7 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & \frac{1}{3} \end{bmatrix}$.

Write down the eigenvalues of A^{-2} and their multiplicities.

Answer: $\frac{1}{25}$ with multiplicity 2; $\frac{1}{4}$ with multiplicity 1; 9 with multiplicity 1.

7. **[2 points]** Let A, B, and C be 3×2 matrices such that $C = -A + 3B$. Also let $\mathbf{v} \in \mathbb{R}^2$ be a vector such that

$$A\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} \quad \text{and} \quad B\mathbf{v} = \begin{bmatrix} -3 \\ 5 \\ -1 \end{bmatrix}.$$

Calculate $C\mathbf{v}$.

Answer: $C\mathbf{v} = \begin{bmatrix} -10 \\ 16 \\ -6 \end{bmatrix}$.

8. **[2 points]** Suppose that A is a 5×5 invertible matrix such that -3 is an eigenvalue of A with multiplicity 2. Let I_5 denote the 5×5 identity matrix. For each statement below, write 'T' if it is true, and write 'F' if it is false. You will receive 0.5 points for each correct answer, lose 0.25 points for each incorrect answer, and receive zero points for an answer left blank. You cannot receive a negative score on this question.

F The dimension of the eigenspace corresponding to the eigenvalue -3 is strictly bigger than 2.

F The value of the characteristic polynomial of A at $\lambda = 0$ is equal to 0.

F The matrix $A^{-1} + \frac{1}{3}I_5$ is invertible.

T $\det(A^T + 3I_5) = 0$.

9. **[1 point]** Let

$$A = \begin{bmatrix} -3 & 4 & \frac{1}{3} & -3 & -2 & 7 & -4 \\ 0 & -3 & -1 & -4 & 11 & -\frac{1}{2} & 2 \\ 0 & 0 & 0 & 2 & -1 & 4 & 1 \\ 0 & 0 & 0 & 2 & -1 & 4 & 1 \end{bmatrix}$$

Write down the dimension of the subspace $W = \left\{\mathbf{b} \in \mathbb{R}^4 \ : \ \text{The linear system } A\mathbf{x} = \mathbf{b} \text{ is consistent}\right\}$.

Solution: $\dim W = 3$.

10. **[1 point]** Determine the value of the parameter t such that the vector $\begin{bmatrix} t \\ -2 \end{bmatrix}$ is an eigenvector of the matrix

$$A = \begin{bmatrix} i & -i-1 \\ -\frac{1}{2} + \frac{1}{2}i & i \end{bmatrix}$$

corresponding to the eigenvalue $\lambda = 1 + i$. You should write your final answer in the form $t = a + bi$ for $a, b \in \mathbb{R}$.

Answer: $t = 2 + 2i$.

11. **[2 points]** For each of the following subsets of $\mathbb{R}^3$, write 'Y' if the set is a *subspace* of $\mathbb{R}^3$, and write 'N' if it is not. You will receive 0.5 points for each correct answer, lose 0.25 points for each incorrect answer, and receive zero points for an answer left blank. You cannot receive a negative score on this question.

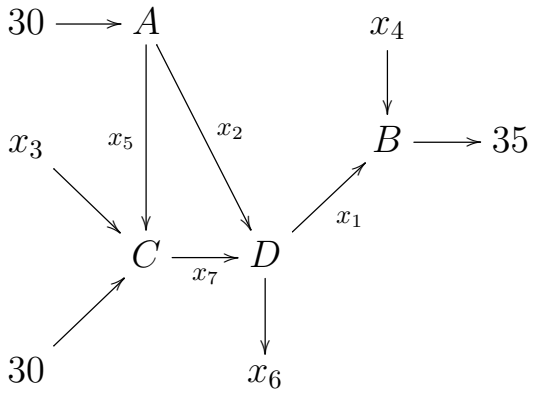
Y The set containing all vectors $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ in $\mathbb{R}^3$ such that $\begin{bmatrix} 3 & -1 & -4 \\ -1 & 2 & 3 \\ 2 & 1 & -1 \\ 5 & 0 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$.

N $\left\{ \begin{bmatrix} x+y+2z \\ y-2z \\ x-z \end{bmatrix} \;\middle|\; x, y, z \in \mathbb{R} \text{ and } x > 0 \text{ and } z < 0. \right\}$

Y The span of the columns of the matrix $A = \begin{bmatrix} -3 & -1 & \frac{3}{2} & 5 \\ -2 & 1 & -\frac{3}{2} & 4 \\ -1 & -2 & 3 & 1 \end{bmatrix}$.

Y $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \;\middle|\; x, y, z \in \mathbb{R} \text{ and } \begin{bmatrix} -3 & 0 & -1 \\ 3 & 1 & 6 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2x \\ -2y \\ -2z \end{bmatrix} \right\}$.

12. **[2 points]** Consider the traffic flow described by the following diagram. The letters A through D label intersections. The arrows indicate the direction of flow (all roads are one-way) and their labels indicate flow in cars per minute.



Write down a linear system describing the traffic flow, i.e., all constraints on the variables $x_i, i = 1, \ldots, 7$. (Do not solve the linear system.)

Solution:

$$\begin{cases} A: & 30 = x_5 + x_2 \\ B: & x_1 + x_4 = 35 \\ C: & x_3 + 30 + x_5 = x_7 \\ D: & x_2 + x_7 = x_1 + x_6 \\ \text{Total}: & 30 + x_3 + 30 + x_4 = 35 + x_6 \end{cases}$$

13. **[2 points]** Let A and B be $n \times n$ invertible matrices. Solve the matrix equation $A(X + B^T) = -2BA$ for the matrix X.

Answer: $X = -2A^{-1}BA - B^T$.

14. **[2 points]** For each of the following matrices, write 'Y' if it is diagonalizable, and write 'N' if it is not. You will receive 1 point for each correct answer, lose 0.5 points for each incorrect answer, and receive zero points for an answer left blank. You cannot receive a negative score on this question.

Y $A = \begin{bmatrix} -2 & 0 & 0 & 0 \\ -4 & 3 & 0 & 0 \\ -1 & 3 & -5 & 0 \\ 0 & -4 & 3 & 2 \end{bmatrix}$

N $B = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$.

Part B: Long Answer Questions

For Questions 15–21, you must show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.

15. [**4 points**] Write down the general solution of the following vector equation in vector parametric form.

$$x_1\begin{bmatrix}1\\0\\3\end{bmatrix}+x_2\begin{bmatrix}-3\\0\\-9\end{bmatrix}+x_3\begin{bmatrix}1\\1\\3\end{bmatrix}+x_4\begin{bmatrix}0\\2\\2\end{bmatrix}+x_5\begin{bmatrix}1\\3\\1\end{bmatrix}=\begin{bmatrix}-3\\0\\-7\end{bmatrix}$$

Solution: We reduce the augmented matrix of the system to the REF:

$$\left[\begin{array}{ccccc|c}1&-3&1&0&1&-3\\0&0&1&2&3&0\\3&-9&3&2&1&-7\end{array}\right]\xrightarrow{R_3\to R_3-3R_1}\left[\begin{array}{ccccc|c}1&-3&1&0&1&-3\\0&0&1&2&3&0\\0&0&0&2&-2&2\end{array}\right]$$

$$\xrightarrow{R_3\to\frac{1}{2}R_3}\left[\begin{array}{ccccc|c}1&-3&1&0&1&-3\\0&0&1&2&3&0\\0&0&0&1&-1&1\end{array}\right]$$

$$\xrightarrow{R_2\to R_2-2R_3}\left[\begin{array}{ccccc|c}1&-3&1&0&1&-3\\0&0&1&0&5&-2\\0&0&0&1&-1&1\end{array}\right]$$

$$\xrightarrow{R_1\to R_1-R_2}\left[\begin{array}{ccccc|c}1&-3&0&0&-4&-1\\0&0&1&0&5&-2\\0&0&0&1&-1&1\end{array}\right]$$

The variables x_1, x_3, x_4 are basic, and x_2, x_5 are free. The general solution is

$$\begin{cases}x_1=3x_2+4x_5-1\\x_2=\text{free}\\x_3=-5x_5-2\\x_4=x_5+1\\x_5=\text{free}\end{cases}$$

The vector parametric form of the solution is

$$\begin{bmatrix}x_1\\x_2\\x_3\\x_4\\x_5\end{bmatrix}=\begin{bmatrix}3x_2+4x_5-1\\x_2\\-5x_5-2\\x_5+1\\x_5\end{bmatrix}=x_2\begin{bmatrix}3\\1\\0\\0\\0\end{bmatrix}+x_5\begin{bmatrix}4\\0\\-5\\1\\1\end{bmatrix}+\begin{bmatrix}-1\\0\\-2\\1\\0\end{bmatrix}$$

16. [**4 points**] Calculate the determinant of the following matrix using the method of co-factor expansion.

$$M = \begin{bmatrix} 1 & 0 & 0 & -5 \\ -7 & -4 & 5 & 11 \\ 2 & 0 & 15 & -3 \\ 1 & 0 & -1 & -6 \end{bmatrix}.$$

Solution: Expanding with respect to the 2nd column, and then again with respect to the 2nd column of the resulting 3×3 matrix, we obtain

$$\begin{vmatrix} 1 & 0 & 0 & -5 \\ -7 & -4 & 5 & 11 \\ 2 & 0 & 15 & -3 \\ 1 & 0 & -1 & -6 \end{vmatrix} = (-4) \begin{vmatrix} 1 & 0 & -5 \\ 2 & 15 & -3 \\ 1 & -1 & -6 \end{vmatrix} = (-4)(15) \begin{vmatrix} 1 & -5 \\ 1 & -6 \end{vmatrix} - (-4)(-1) \begin{vmatrix} 1 & -5 \\ 2 & -3 \end{vmatrix}$$

$$= (-4)(15)(-1) - 28 = 60 - 28 = 32.$$

17. [**4 points**] Consider the matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 3 & -2 & 0 \\ 5 & -1 & -4 \end{bmatrix}$. Calculate the eigenvalues of A.

Solution: Starting with an expansion with respect to the third column, we have

$$\det(A - \lambda I) = \begin{vmatrix} -1-\lambda & 2 & 0 \\ 3 & -2-\lambda & 0 \\ 5 & -1 & -4-\lambda \end{vmatrix} = (-4-\lambda)\begin{vmatrix} -1-\lambda & 2 \\ 3 & -2-\lambda \end{vmatrix} = (-4-\lambda)(\lambda^2+3\lambda-4).$$

The roots of $(\lambda^2 + 3\lambda - 4)$ are

$$\lambda = \frac{-3 \pm \sqrt{(3)^2 - 4(-4)}}{2} = \begin{cases} 1 \\ -4 \end{cases}$$

Thus, the eigenvalues of A are -4 and 1.

18. **[4 points]** Let $A = \begin{bmatrix} -2 & -1 & 4 \\ -1 & -2 & -4 \\ 2 & -2 & 5 \end{bmatrix}$. Find a basis for the eigenspace of A corresponding to the eigenvalue $\lambda = -3$.

Solution: For the eigenvalue $\lambda = -3$, we have to find the basis of $\text{Nul}(A + 3I)$:

$$\left[\begin{array}{c|c} A+3I & 0 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & -1 & 4 & 0 \\ -1 & 1 & -4 & 0 \\ 2 & -2 & 8 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|c} 1 & -1 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Thus, the eigenspace is given by

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 - 4x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix}$$

and a basis of this eigenspace is

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

19. The pacific island of Nukufeta has two local bus companies: *Coachfeta* and *Malamus*. Each of the senior residents of the island has received the discount membership plan of one of the two companies (but not both). Data analysis results released in the beginning of 2018 showed that $\frac{1}{3}$ of the senior residents had *Coachfeta*'s membership plan, while the remaining $\frac{2}{3}$ of the senior residents had *Malamus*'s membership plan.

In the course of 3 months of competitive marketing, *Coachfeta* lost $\frac{1}{5}$ of its senior members to *Malamus*, and *Malamus* lost $\frac{2}{5}$ of its senior members to *Coachfeta*.

(a) **[1 point]** Write down the migration matrix M and the initial state vector $\mathbf{x}_0$ for this Markov chain.

Solution:

$$M = \begin{bmatrix} \frac{4}{5} & \frac{2}{5} \\ \frac{1}{5} & \frac{3}{5} \end{bmatrix}, \quad \vec{x}_0 = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}$$

Another possible migration matrix (if one reverses the order of the companies) is

$$M = \begin{bmatrix} \frac{3}{5} & \frac{1}{5} \\ \frac{2}{5} & \frac{4}{5} \end{bmatrix}, \quad \vec{x}_0 = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$$

(b) **[1 point]** At the end of March 2018, what fraction of the senior residents had the membership of *Coachfeta*? What fraction of the senior residents had the membership of *Malamus*?

Solution: $\vec{x}_1 = M\vec{x}_0 = \begin{bmatrix} \frac{8}{15} \\ \frac{7}{15} \end{bmatrix}$

(c) **[4 points]** Suppose that the first quarter's migration trend continues for several successive quarters. Then in the long run what percentage of senior residents will possess the discount membership plan of each of the two companies? You should clearly indicate the fraction of members for each company.

Solution: To find the steady-state vector, we must find an eigenvector of eigenvalue 1. We row reduce:

$$\left[\begin{array}{c|c} M - I & 0 \end{array}\right] = \left[\begin{array}{cc|c} -\frac{1}{5} & \frac{2}{5} & 0 \\ \frac{1}{5} & -\frac{2}{5} & 0 \end{array}\right] \xrightarrow[R_2 \to 5R_2]{R_1 \to -5R_1} \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 1 & -2 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

The general solution is thus

$$\vec{x} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

We choose the value of the free variable so that the sum of the entries of $\vec{x}$ is equal to one:

$$(2+1)\, x_2 = 1 \implies x_2 = \frac{1}{3}.$$

Thus the steady-state vector is

$$\vec{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}.$$

Since M is regular stochastic, the long term behaviour is given by the steady-state vector. Thus, in the long term, $\frac{2}{3}$ of the total number of passengers will possess the discount membership plan of *Coachfeta*, and $\frac{1}{3}$ of passengers will possess the discount membership plan of *Malamus*.

20. [**4 points**] Let

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 1 & 2 & 9 \\ 3 & 9 & 11 \end{bmatrix}$$

Find the inverse of A.

Solution: We row reduce the super-augmented matrix:

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 4 & 1 & 0 & 0 \\ 1 & 2 & 9 & 0 & 1 & 0 \\ 3 & 9 & 11 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|ccc} 1 & 3 & 4 & 1 & 0 & 0 \\ 0 & -1 & 5 & -1 & 1 & 0 \\ 0 & 0 & -1 & -3 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to -R_3} \left[\begin{array}{ccc|ccc} 1 & 3 & 4 & 1 & 0 & 0 \\ 0 & -1 & 5 & -1 & 1 & 0 \\ 0 & 0 & 1 & 3 & 0 & -1 \end{array}\right] \xrightarrow[R_1 \to R_1 - 4R_3]{R_2 \to R_2 - 5R_3} \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & -11 & 0 & 4 \\ 0 & -1 & 0 & -16 & 1 & 5 \\ 0 & 0 & 1 & 3 & 0 & -1 \end{array}\right]$$

$$\xrightarrow{R_2 \to -R_2} \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & -11 & 0 & 4 \\ 0 & 1 & 0 & 16 & -1 & -5 \\ 0 & 0 & 1 & 3 & 0 & -1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 3R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -59 & 3 & 19 \\ 0 & 1 & 0 & 16 & -1 & -5 \\ 0 & 0 & 1 & 3 & 0 & -1 \end{array}\right]$$

Thus A is invertible, and its inverse is

$$A^{-1} = \begin{bmatrix} -59 & 3 & 19 \\ 16 & -1 & -5 \\ 3 & 0 & -1 \end{bmatrix}.$$

21. An economy consists of two sectors: Agriculture and Service. In order to produce one unit, the Agriculture sector consumes $\frac{1}{2}$ units from Agriculture and $\frac{3}{10}$ units from Service. Also, in order to produce one unit, the Service sector consumes $\frac{1}{2}$ units from Agriculture and $\frac{1}{5}$ units from Service.

(a) **[1 point]** Write the Consumption Matrix for this economy.

Solution:

$$C = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{3}{10} & \frac{1}{5} \end{bmatrix}$$

(b) **[1 point]** Write the Leontief Input-Output Model production equation.

Solution: The equation is $\vec{x} = C\vec{x} + \vec{d}$, or $(I - C)\vec{x} = \vec{d}$.

(c) **[3 points]** Determine the production levels needed to satisfy a final demand of 5 units from the Agriculture sector and 3 units from the Service sector.

Solution: We solve the equation $A\vec{x} = \vec{d}$, where

$$A = I_2 - C = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{3}{10} & \frac{4}{5} \end{bmatrix}, \quad \vec{d} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}.$$

We have

$$\det A = \frac{4}{10} - \frac{3}{20} = \frac{5}{20} = \frac{1}{4} \neq 0.$$

Thus A is invertible and

$$A^{-1} = 4 \begin{bmatrix} \frac{4}{5} & \frac{1}{2} \\ \frac{3}{10} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{16}{5} & 2 \\ \frac{6}{5} & 2 \end{bmatrix}.$$

Therefore,

$$\vec{x} = A^{-1}\vec{d} = \begin{bmatrix} \frac{16}{5} & 2 \\ \frac{6}{5} & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 22 \\ 12 \end{bmatrix}.$$

So the Agriculture sector needs to produce 22 units and the Service sector needs to produce 12 units.